

Brad McKellar

Provo, UT 84606

portfolio: [bmckellar1.github.io](https://github.com/bmckellar1)

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Education

Bachelor of Science, Mechanical Engineering

Brigham Young University—Provo, Utah

Expected Apr. 2028

- Focus on robotics and medical devices

Experience

Research Assistant

Brigham Young University, College of Engineering, Terry Research Lab

May 2026–Present

- Redesigned mechanical components to improve manufacturability, yield and robustness of a novel target medication delivery mechanism
- Designed and executed comprehensive *in vitro* test of delivery system
- Analyzed data in Python and Excel; presented findings at a university sponsored research conference
- Designed an *in vivo* test to validate delivery system in a living creature

Engineering Intern

Foldax Inc., Salt Lake City, Utah

May 2025–August 2025

- Designed components and subsystems to improve manufacturing process yield
- Developed Python prediction software to guide manufacturing decisions and reduce material waste
- Wrote test protocols, result reports, and documentation to support FDA and patent compliance
- Updated official assemblies and drawings using SolidWorks PDM

Prototyping Lab Assistant

Brigham Young University College of Engineering, Prototyping Lab, Provo Utah

April 2024 –August 2026

- Collaborated with faculty to design and manufacture components for research projects
- Programmed and machined complex prototypes using Haas CNC machines
- Trained and assisted students in prototyping entrepreneurial, personal, and capstone projects

Skills

Programming and Robotics

- Python, C++, HTML
- Data analysis in Python and Excel
- ROS, MuJoCo, and Stable-Baselines3
- Built and deployed custom applications on a home server
- Custom PCB design, prototyped circuitry and filtering
- Embedded system design using PIC, Arduino, ESP32, and Raspberry Pi

Mechanical Design and Manufacturing

- SolidWorks (including PDM), Fusion360, and Onshape
- Mastercam with Haas CNC mills and lathes, including 4 axis
- 3D printing, FDM and SLA